**Cluster: ABELL 0267**
**ObsID: 1448**


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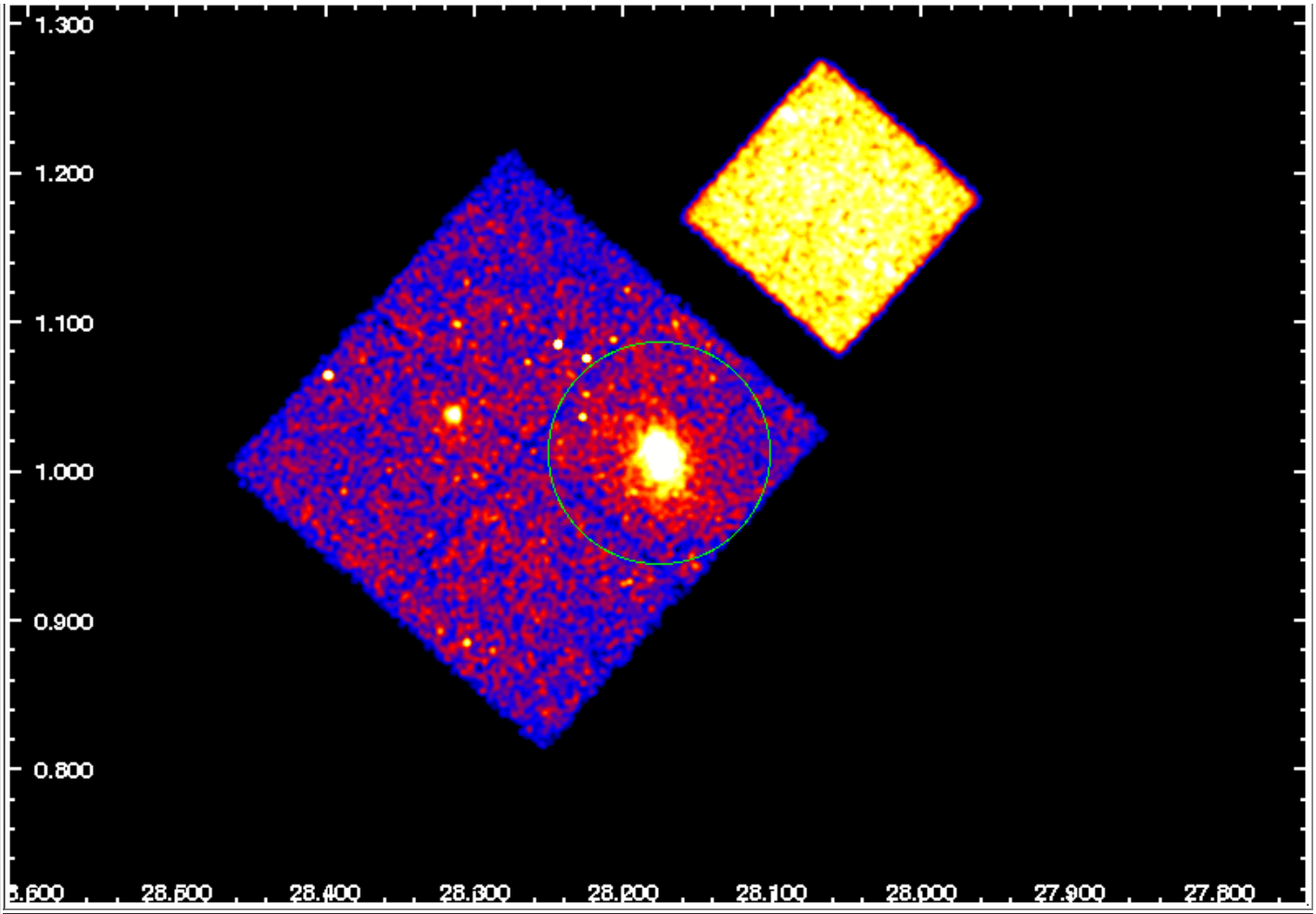
Col. **(1)** Exposure time; col. **(2)** right ascension of cluster center; col. **(3)** declination of cluster center; col. **(4)** Galactic absorbing column density, $N_H$; col. **(5)** redshift; col. **(6)** maximum radius from cluster center, $R_{max}$; col. **(7)** minimum counts used per $T_X$ annulus; col. **(8)** average cluster temperature; col. **(9)** average cluster metal abundance; and col. **(10)** bolometric luminosity;

| Exp Time | R.A. | Decl. | $N_H$ | $z$ | $R_{max}$ | Min. cts. | $T_X$ | Abund. | $L_{bol}$ |
|---|---|---|---|---|---|---|---|---|---|
| ksec | hr:min:sec | deg:min:sec | $10^{20}$ cm$^{-2}$ | --- | pixels | counts | keV | Solar | $10^{44}$ ergs/s |
| (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| 7.9 | 01:52:42.269 | +01:00:45.33 | 2.74 | 0.2300 | 540 | 5000 | 6.79 | 0.26 | 8.700 |

Click here to download a master table of profiles for this cluster.

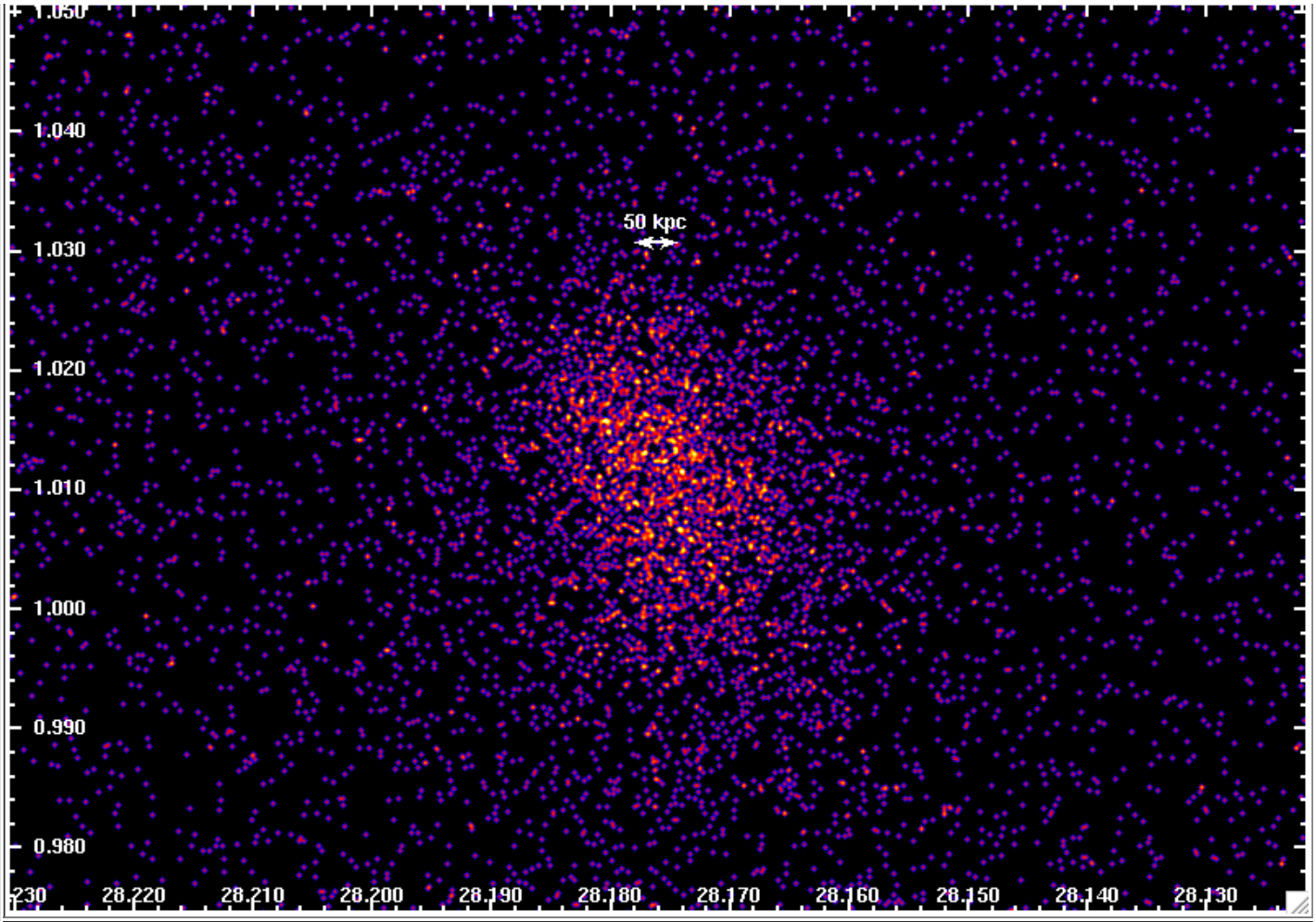
All images generated using SAOImage DS9, developed by the Smithsonian Astrophysical Observatory.

**Smoothed and binnd wide-field Chandra X-ray image of cluster**



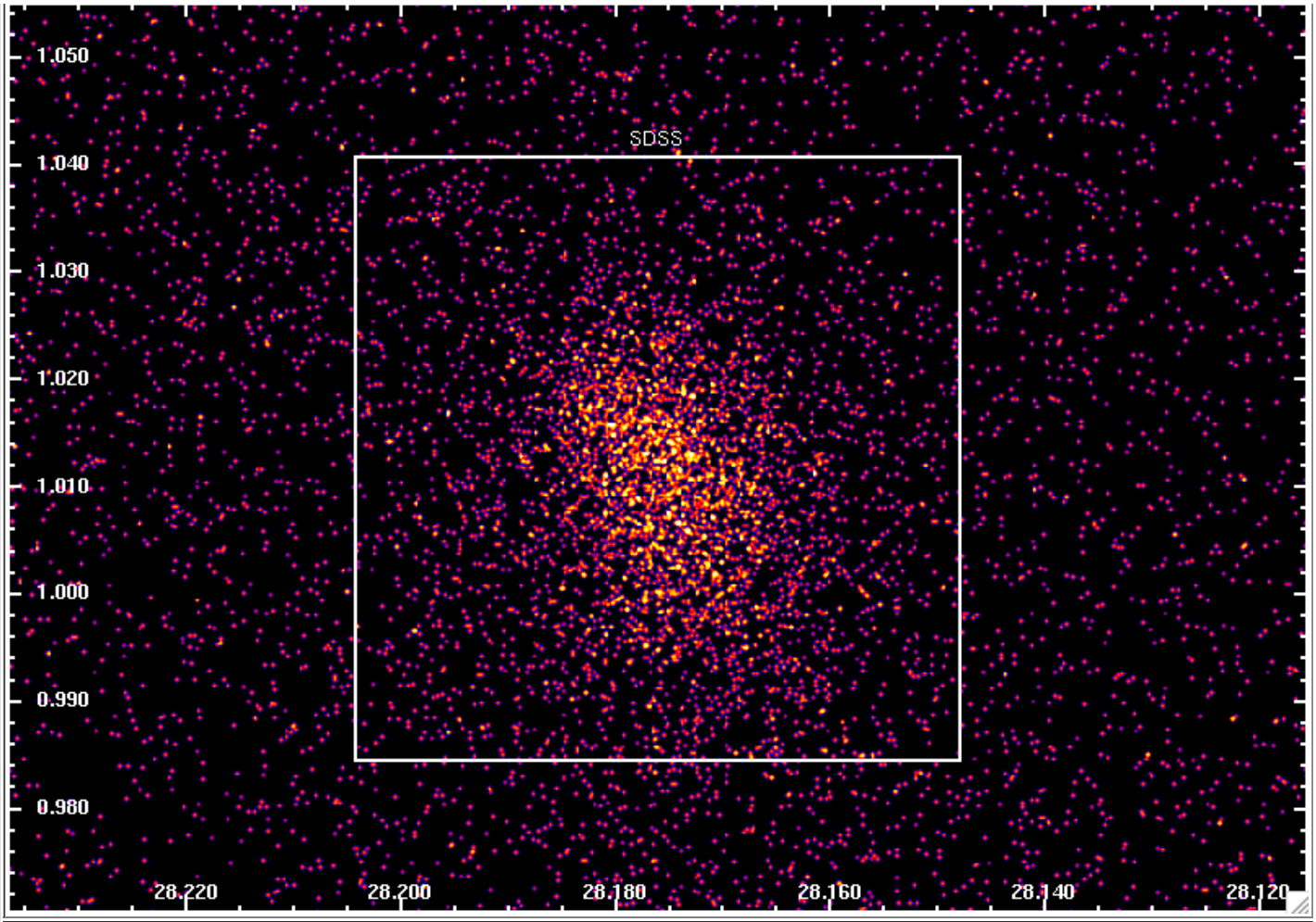

A gaussian smoothing kernel with size 2 was applied to the flare-cleaned, level-2 events file binned with a factor of 8 to produce this image. Point sources have not been excluded and no exposure correction has been applied to this image.

**Smoothed and binned Chandra X-ray image of the cluster "core"**




Same as the wide-field image above but the binning factor used here was 2.

**SDSS/DSS footprint overlaid on cluster core**




The SDSS/DSS footprint (white box) is centered on the cluster core.

**Sloan Digital Sky Survey (SDSS) image of cluster core**

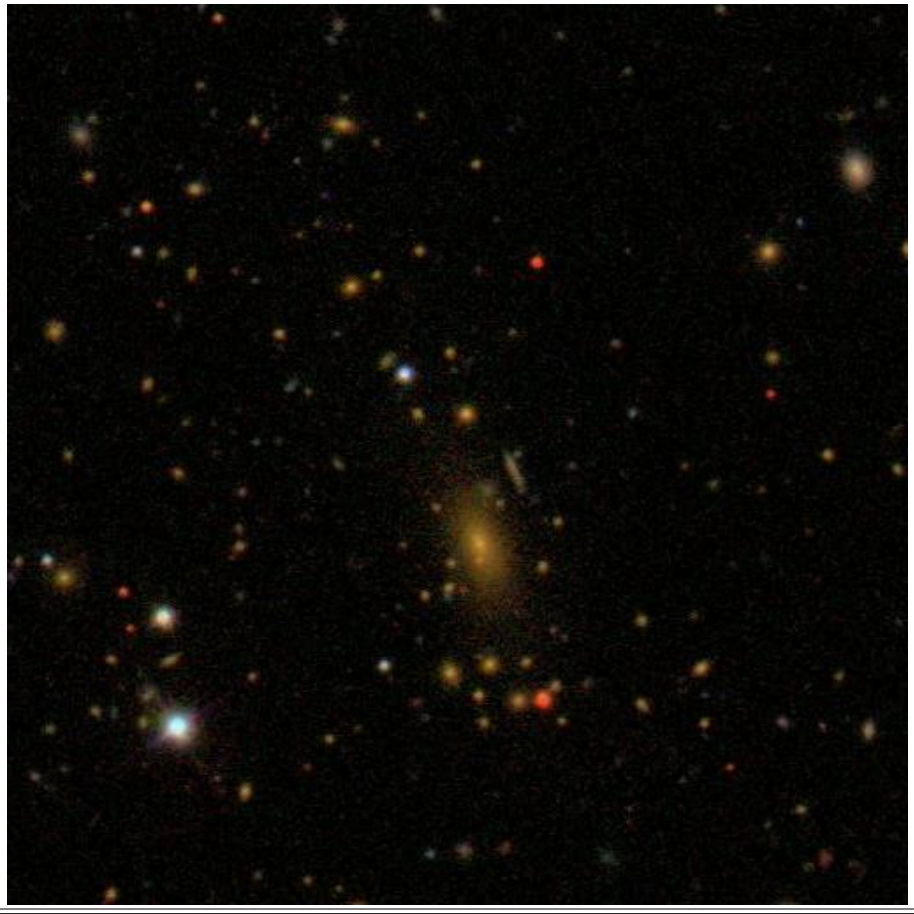
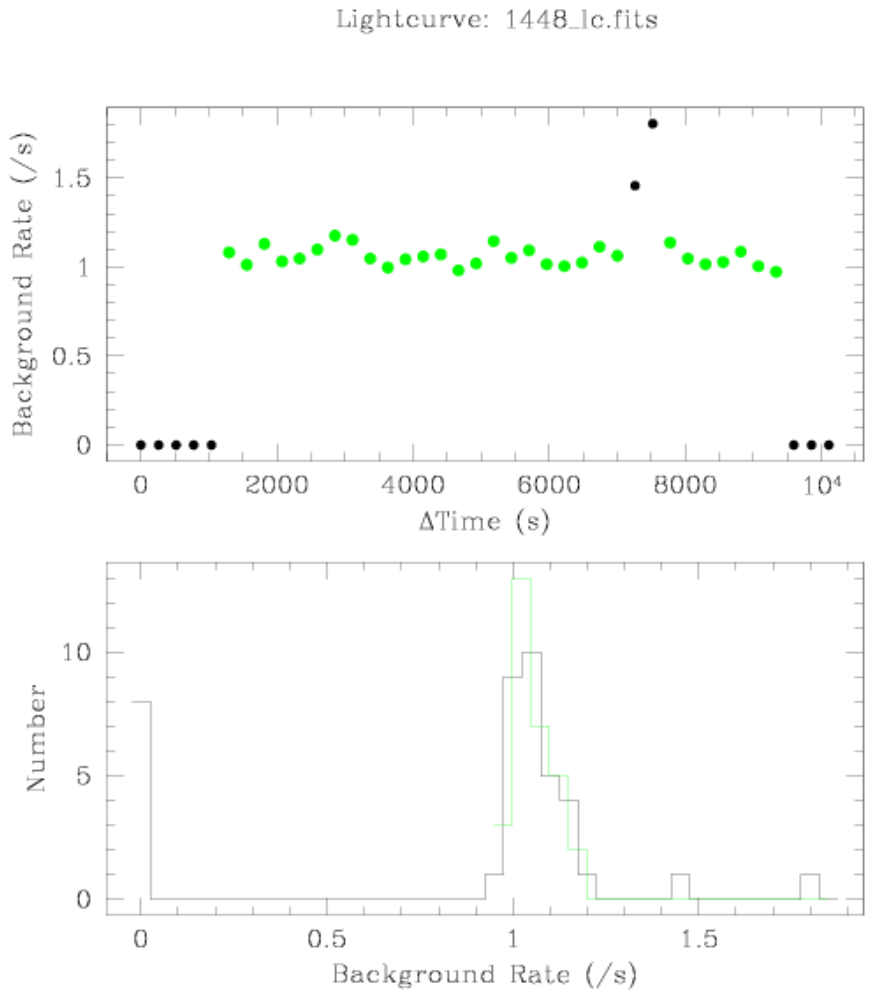


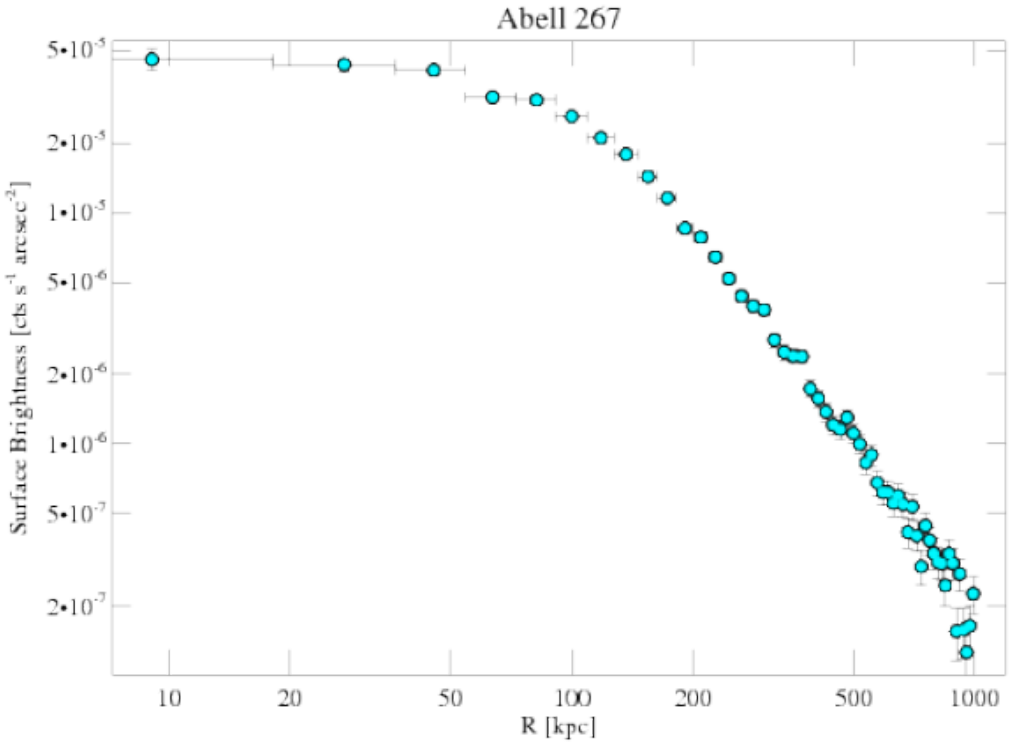
Image taken from Sloan Digital Sky Survey (SDSS)

**Background light curve**



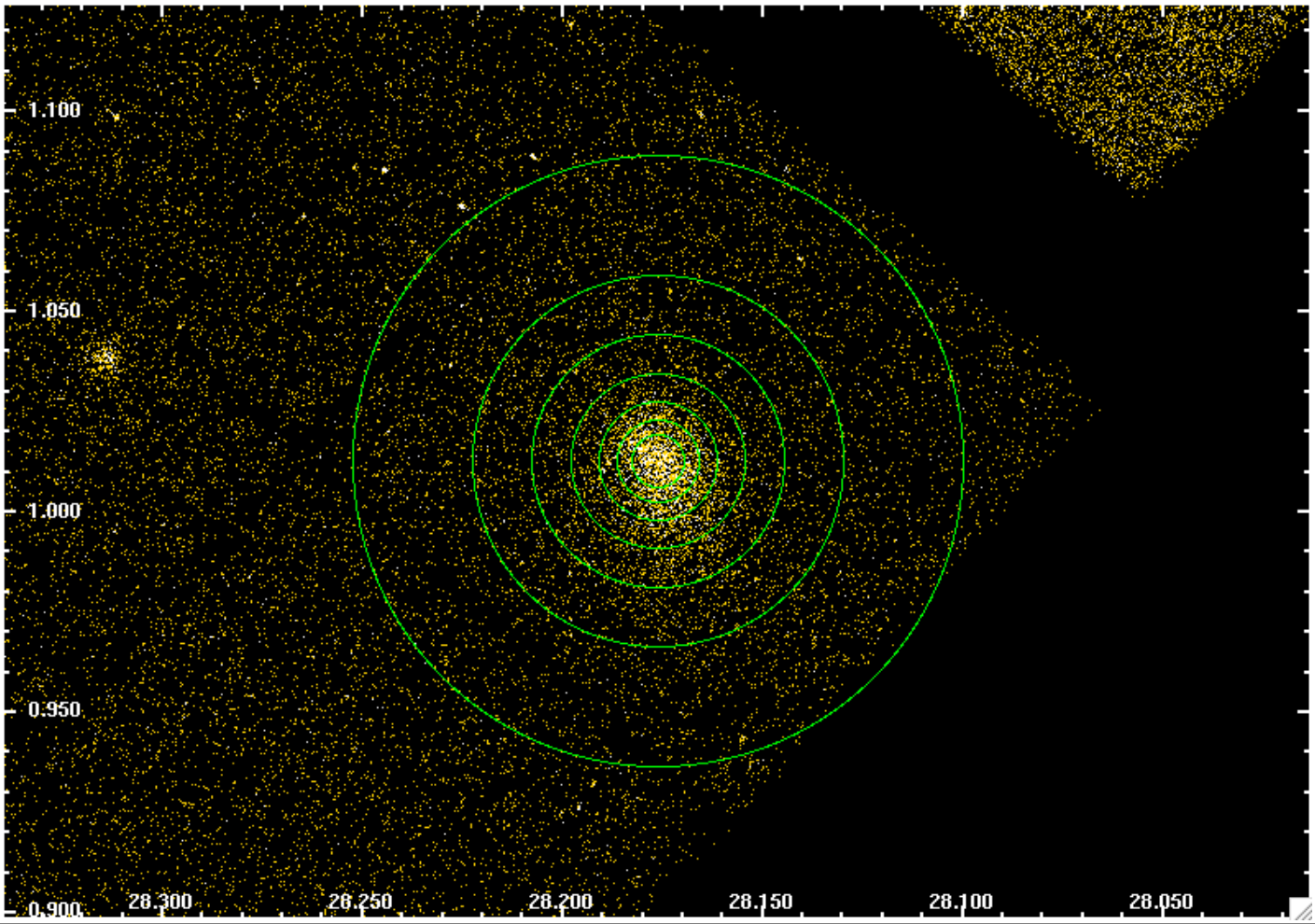

To check for contamination from background flares or periods of excessively high background, light curve analysis was performed using Maxim Markevitch's contributed CIAO script lc_clean.sl. Periods with count rates >= 3sigma and/or a factor >= 1.2 of the mean background level of the observation were removed from the GTI file.

**Surface brightness profile**




We began by extracting surface brightness profiles from the 0.7-2.0 keV energy range using concentric annular bins of size 5" (~10 ACIS pixels) originating from the X-ray center. To remove the effects of vignetting and exposure time fluctuations we corrected each surface brightness profile by an observation specific normalized radial exposure profile.

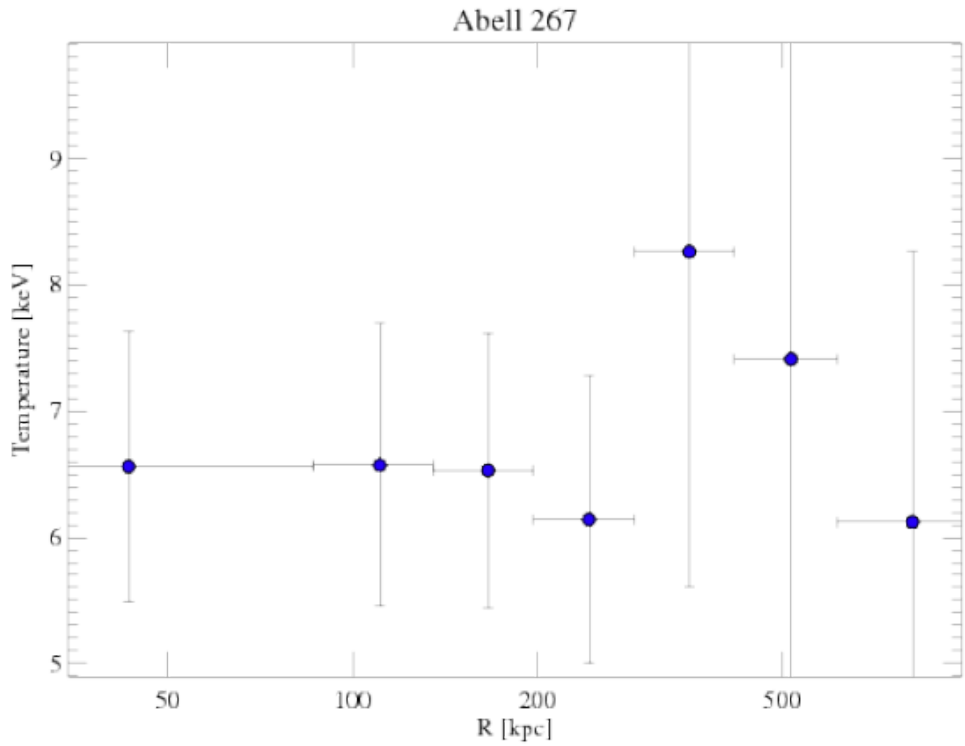
**Annuli used for generating temperature profile**




Each annulus is centered at RA: 01:52:42.269, Decl: +01:00:45.33 and contains 5000 counts. Links to an image of each spectrum and the corresponding best-fit MeKaL model (solid-line) are provided below the image.

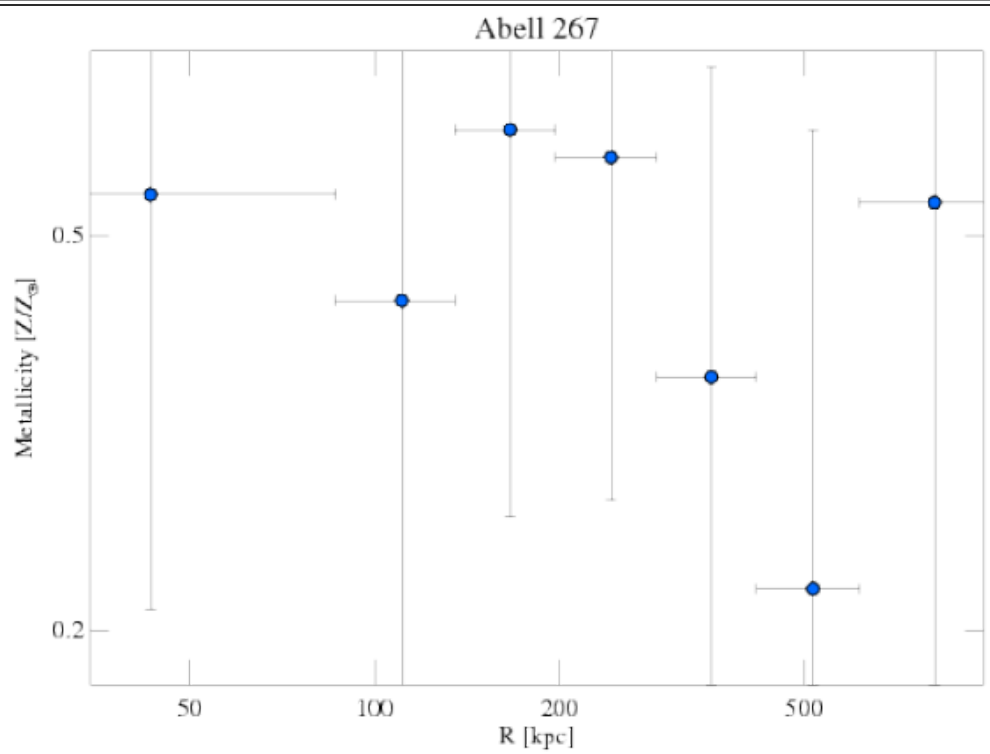
| Links to images of spectra | | | | |
|---|---|---|---|---|
| Annulus 1 | Annulus 2 | Annulus 3 | Annulus 4 | Annulus 5 |
| Annulus 6 | Annulus 7 | | | |

**Temperature profile**



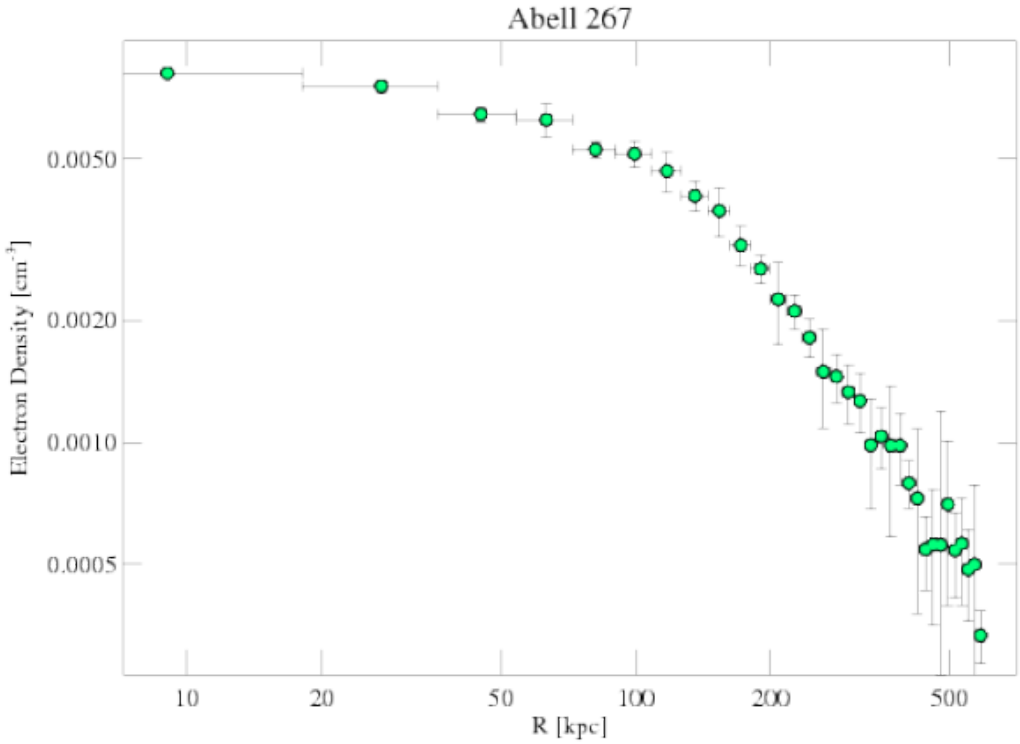

The temperature profile corresponding to the image and annuli shown above. Uncertainties are 90% confidence.

**Metal abundance profile**



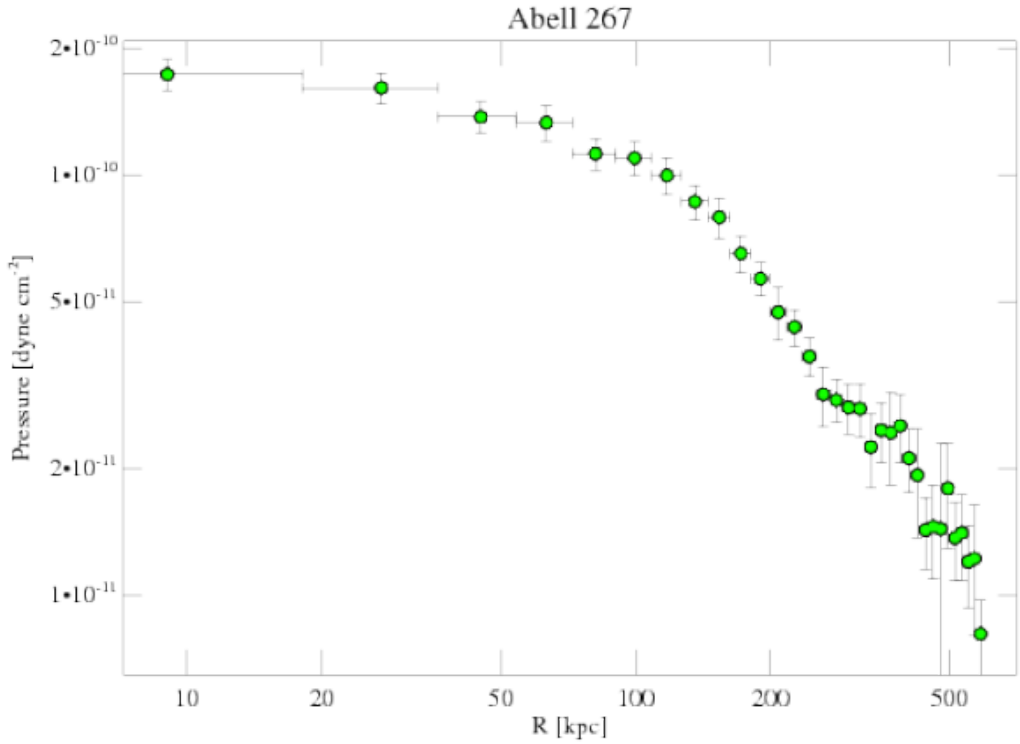

The abundance profile corresponding to the image and annuli shown above. Uncertainties are 90% confidence.

**Density profile**



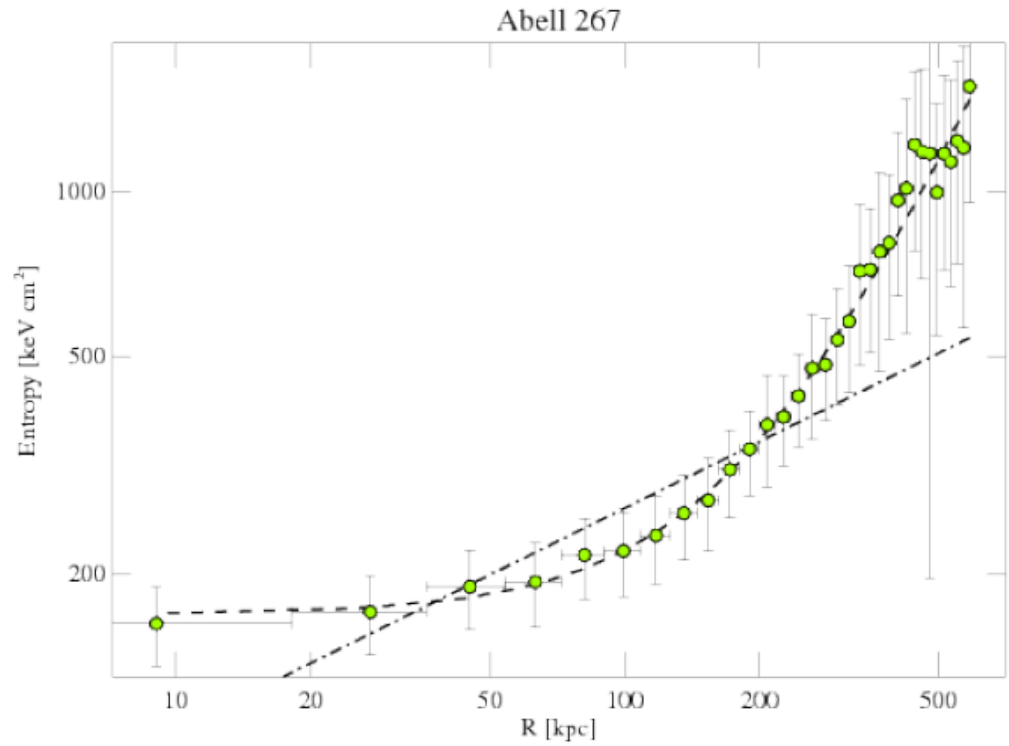

Electron gas density profile derived from deprojecting the surface brightness with spectral information.

**Pressure profile**




Electron gas pressure profile derived from the density and temperature profiles.

**Entropy profile**




Entropy profile derived from density and temperature as $K=Tn_e^{-2/3}$. The dashed line is the best-fit for the model $K(r) = K_0+K_{100}(r/100\ kpc)^{\alpha}$ with fit values:

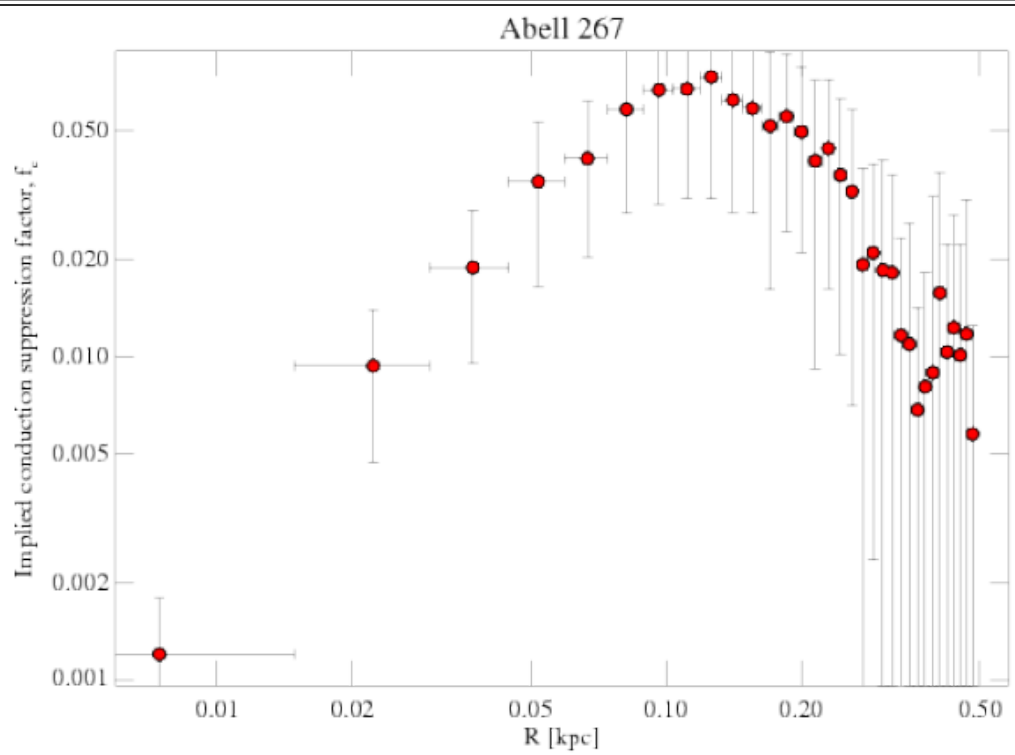
$K_0$ = 168.56 ± 17.64 keV cm$^2$

$K_{100}$ = 51.81 ± 21.05 keV cm$^2$

$\alpha$ = 1.82 ± 0.38.

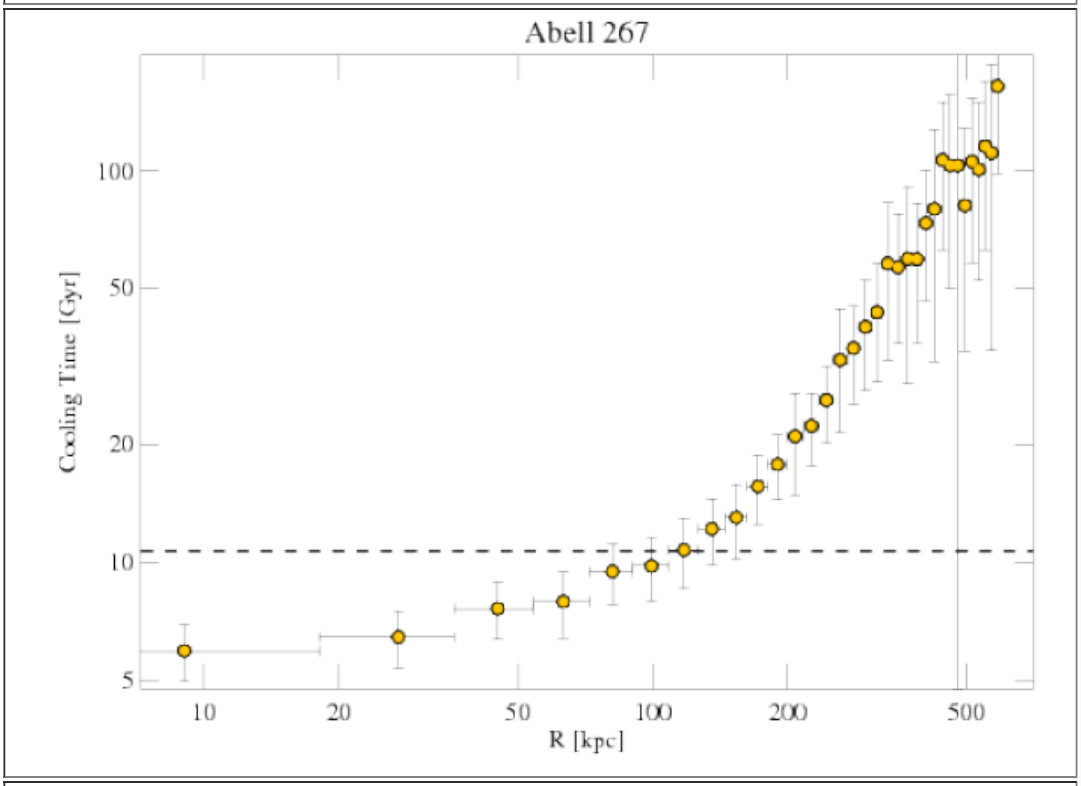
The dashed-dotted line is the best-fit for the model $K=K_{100}(r/100\ kpc)^{\alpha}$.

**Conduction suppresion profile**



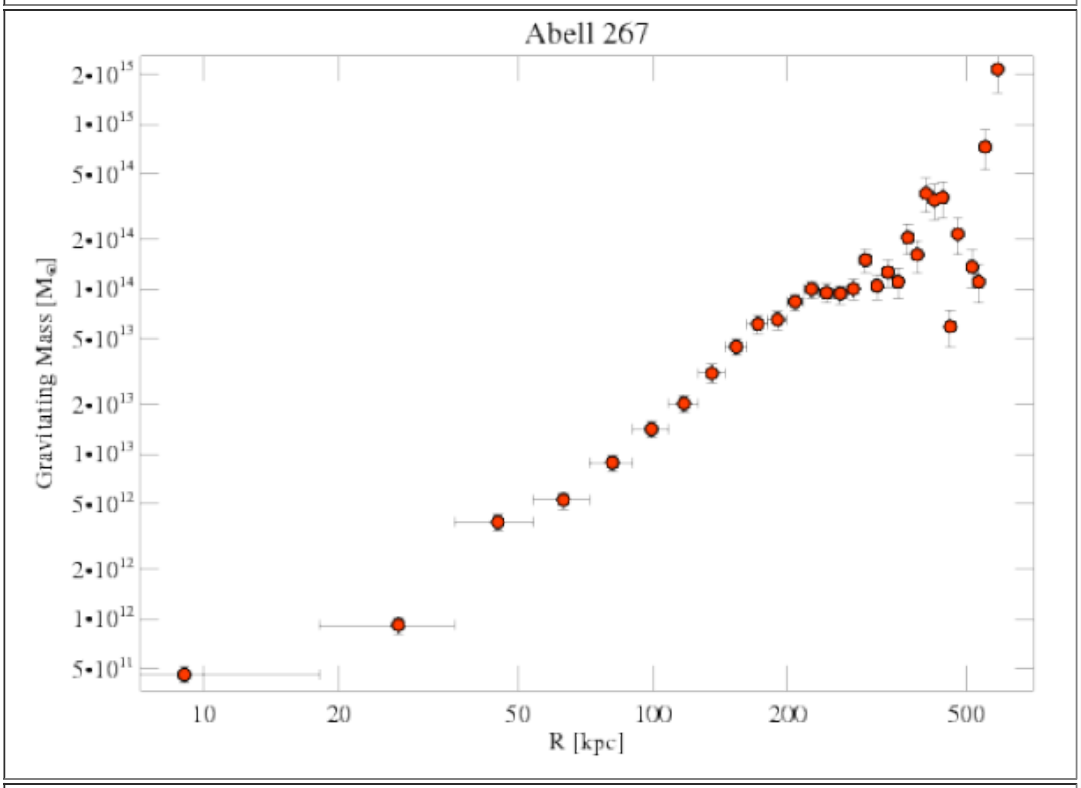

Conduction suppression factor, $f_c$ profile derived from the entropy profile assuming Spitzer conduction.

**Cooling time profile**




Gas cooling time profile using the formulation $t_{cool}=(3/2\ nkT)/(n_e n_p \Lambda(T,Z))$ for isochoric cooling. The dashed horizontal line represents the age of the Universe at the redshift of the cluster.

**Gravitating mass profile**




Gravitating mass profile derived from assuming hydrostatic equilibrium and differentiating the log-space electron density profile with a piecewise method. The d(ln T)/dr term has been assumed to be negligible

(a terrible assumption). **CAUTION** THIS WAS NOT THE FOCUS OF OUR WORK, USE AT YOUR OWN RISK!

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